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ORIGINAL RESEARCH

Enhancing legal assistance with AI: a comprehensive approach to intent classification and domain specific model tuning

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Abstract

Our research explores the application of Large Language Models for enhancing legal advice through AI-driven conversational agents, focusing on the analysis and interpretation of user-generated content from a prominent online legal community. By leveraging BERTopic for topic modeling and GPT-3.5 for labeling of user intents, we constructed a dataset that served as the foundation for fine-tuning the Mistral 7B language model. This process involved Supervised Fine Tuning and Direct Preference Optimization, culminating in the creation of a Domain Specific Language Model for legal expertise. This model was then employed to guide GPT-3.5 responses, ensuring domain-specific relevance and contextual appropriateness. The evaluation of these responses was conducted using GPT-4, supplemented by human assessment, to gauge the quality of the models' outputs. We leverage an innovative end-to-end pipeline, building upon the framework proposed by Wang et al. (Empower large language model to perform better on industrial domain-specific question answering, 2023. arXiv:2305.11541), specifically adapted for the legal domain. This pipeline provides a foundational framework for developing sophisticated and scalable user-facing chatbots, presenting an alternative to the more common retrieval augmented generation techniques. Our contributions include providing datasets for further research on Legal Question Answering systems and advancements in human alignment of responses from chatbots. This research advances the technical frontier in AI-driven legal assistance and opens new avenues for enhancing the accessibility and quality of legal support through conversational AI technologies.

Keywords Legal question answering · Chatbots · Direct preference optimization · Mistral 7B · GPT4 · BERTopic

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1 Introduction

The intersection of Generative Artificial Intelligence and the legal domain is heralding a transformative era in the delivery and accessibility of legal services. With the advent of Large Language Models (LLMs), the landscape of legal technologies is on the verge of unprecedented advancements, promising sophisticated and cost-effective solutions for a myriad of legal tasks. From judgment summarization to emulating the nuanced reasoning of seasoned lawyers, proprietary LLMs like GPT3.5 and GPT4 and open-source models like Llama and Mistral have the potential to significantly augment the efficiency and accuracy of legal operations, supporting legal professionals in their complex decision-making processes (Deroy et al. 2023; Vats et al. 2023; Blair-Stanek et al. 2023; Yu et al. 2022).

Parallelly, the Legal Question Answering (LQA) systems are emerging as reliable tools for enhancing and democratizing access to legal assistance. By leveraging the prowess of LLMs, these AI-driven platforms, including legal chatbots and virtual assistants, have the potential to bridge the gap between the layperson and the often-inaccessible legal guidance, mitigating traditional barriers of cost and complexity (Abdallah et al. 2023; Queudot et al. 2020; OECD, 2016; Souza and Zarnow 2024). Such innovations not only enhance the public's access to judicial institutions but also underscore the pivotal role of AI in reshaping legal service delivery.

However, the journey towards integrating LLMs within the legal ecosystem is not without its challenges. The prevalence of legal hallucinations and inconsistencies in LLM responses has raised concerns regarding their reliability and accuracy, particularly when tasked with complex legal queries (Dahl et al. 2024; Dash et al. 2023; Ji et al. 2023; Wachter et al. 2024). Addressing these issues necessitates a multifaceted approach, incorporating strategies like domain-specific pretraining, supervised fine-tuning, and innovative model interaction paradigms, as proposed by Wang et al. (2023), to refine the performance of LLMs on domain-specific tasks.

Amidst these technological strides and challenges, online forums and communities have emerged as valuable sources of data for training and enhancing LQA systems. By capturing the diverse nature of legal inquiries in laypersons' language, these platforms offer a unique opportunity to provide insights and data to fine-tune LLMs (Li et al. 2022; Chung et al. 2022; Hu et al. 2021; Wang et al. 2023). The creation of comprehensive datasets from these forums for intent classification, fine-tuning, and human alignment marks a crucial step towards developing more sophisticated and reliable LQA systems. Against this backdrop, our research explores the application of Large Language Models (LLMs) for enhancing legal advice through AI-driven conversational agents, focusing on the analysis and interpretation of user-generated content from a prominent online community. The proposed framework and the methodology are briefly outlined below.

Fig. 1 presents an overall flow of the research methodology adopted by us for enhancing legal assistance by AI-driven chatbots. At the outset, the Reddit

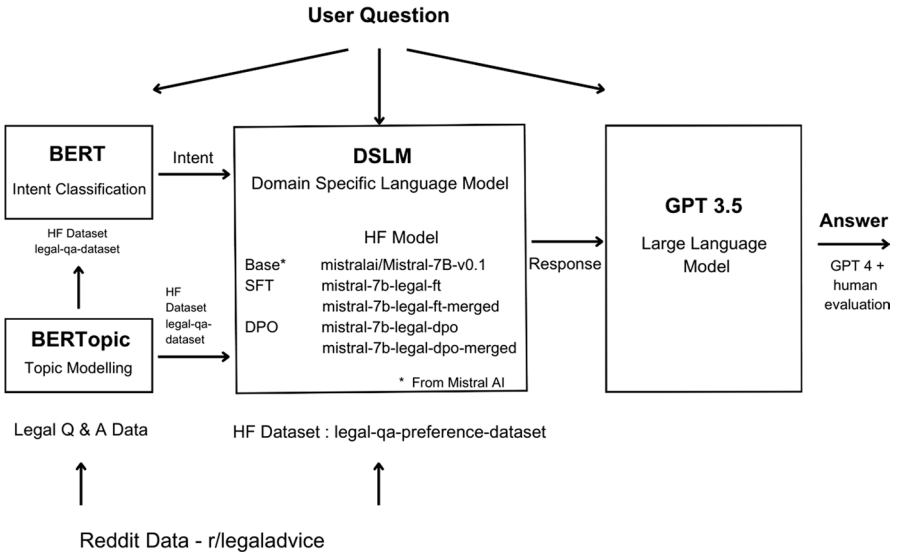

Fig. 1 Proposed legal assistance Q&A framework

community "r/legaladvice" serves as the source for legal Q&A data. This raw data underwent topic modeling using BERTopic (Grootendorst 2022), which along with GPT3.5, facilitated the extraction, classification and labelling of user intents, forming the "legal-qa-dataset". This dataset along with “legal-qa-preference-dataset” became the substrate for fine-tuning the Mistral 7B (Jiang et al. 2023) language model, a process executed in two stages: Supervised Fine Tuning (SFT) and Direct Preference Optimization (DPO). The resultant Domain Specific Language Model (DSLM), fine-tuned for legal expertise, was then utilized to guide the responses of the GPT-3.5 model, ensuring they were contextually relevant and domain-specific.

The final step in the process was the evaluation of the responses. The approach involved using GPT-4, augmented by human evaluation, to assess the quality of the answers generated by the different models. The evaluation code, the datasets, and the fine-tuned models are made available for community access, reproducibility, and research advancement. This model pipeline not only delineates the progression of development phases but also integrates the domain adaptation methodology proposed by Wang et al. (2023), which serves as a foundational framework potentially applicable to the creation of sophisticated legal advice-giving chatbots or other domain-specific user-facing conversational agents.

1.1 Hypotheses

To guide our research and validate the effectiveness of the proposed pipeline, based on the extant literature, we formulated the following hypotheses:

- H1 Transformer-based topic modeling techniques, such as BERTopic, can effectively identify the range of intents behind legal queries sourced from online communities. Furthermore, BERTopic's integration with Large Language Models (LLMs), like GPT3.5, can be used to name and label these identified intents efficiently.
- H2 A small language model, such as BERT (as used in our study), can be trained using intents identified through topic modeling to classify legal queries accurately.
- H3 Publicly available legal queries, responses, and community ratings can be used to fine-tune and improve human alignment in large language models, such as Mistral 7B. This fine-tuning would improve the quality, accuracy, and contextual relevance of the legal responses generated.
- H4 The integration of query intent information, along with DSLM-generated responses, can enhance the quality of responses provided by commercial LLMs, such as GPT3.5 (as used in our study), by improving both contextual relevance and response accuracy.

1.2 Contributions

Our research, by validating the hypotheses, makes the following contributions to the legal technology:

- We successfully demonstrate the use of BERTopic to map the broad range of user intents found within legal queries from a popular online community, such as Reddit. We further show that LLMs like GPT3.5 can be integrated to provide effective and contextually relevant intent names, which can assist in downstream applications of intent classification.
- We validate the use of small language models like BERT to train intent classifiers that can accurately categorize legal queries based on the mined intents, further proving the feasibility of smaller models for targeted legal query classification tasks.
- We show how publicly available legal queries, and their associated responses can be leveraged to fine-tune large language models like Mistral 7B. By employing Direct Preference Optimization (DPO) and other fine-tuning techniques, we enhance the human alignment and contextual accuracy of legal responses generated by LLMs.
- We demonstrate that the integration of query intent information with DSLM-generated responses can significantly improve the quality of responses from commercial LLMs, such as GPT3.5. This integration enhances both the relevance and accuracy of legal advice provided by the system, ensuring more contextually tailored responses for users.
- Additionally, the datasets developed through this research, including the "legal-qa-dataset" and "legal-qa-preference-dataset," are made publicly available. These datasets provide a valuable resource for advancing the evaluation of Legal Question Answering (LQA) systems (Abdallah et al. 2023) and contribute to ongo-

ing research in fine-tuning and human alignment methodologies (Bai et al. 2022; Chung et al. 2022; Rafailov et al. 2024). These resources are intended to support further developments in AI-driven legal assistance tools and LQA systems.

2 Research background

The integration of Artificial Intelligence (AI) into the legal domain has the potential to fundamentally transform the delivery and accessibility of legal services. Recent advancements in AI, especially through the development of Large Language Models (LLMs), are significantly enhancing the capabilities of legal technologies in providing sophisticated solutions for legal research and document analysis and enhancing the efficiency of legal processes. LLMs have shown exceptional utility in legal tasks such as judgment summarization (Deroy et al. 2023), judgment and statute prediction (Vats et al. 2023), statutory reasoning (Blair-Stanek et al. 2023) and emulating legal reasoning akin to a lawyer (Yu et al. 2022). These applications would not only augment the accuracy and efficiency of legal operations but also support legal professionals in their complex tasks. The profound ability of these models to process and interpret extensive legal texts allows for invaluable assistance in legal decision-making and the preparation of legal documents, heralding a new era in the confluence of AI and legal services.

In particular, Legal Question Answering (LQA) systems, empowered by advancements in Large Language Models (LLMs), are reshaping access to legal assistance, making it more accessible to the general public and individual litigants (Abdallah et al. 2023). These AI-driven platforms, such as law chatbots and virtual assistants, leverage natural language processing technology to provide answers to legal inquiries, offering a bridge over the traditional barriers of high legal fees and complex legal jargon. By facilitating affordable and easily accessible legal guidance, LQA systems have the potential to play a crucial role in democratizing legal information. This would be particularly beneficial for laypersons, enabling them to make informed decisions and access judicial institutions with greater ease (Queudot et al. 2020; OECD, 2016). Furthermore, the integration of chatbot interfaces within court services, as explored by Souza and Zarnow (2024), exemplifies the potential of these technologies to further enhance the delivery and accessibility of court services. Through such innovations, LQA systems and chatbots would not just augment the efficiency of legal professionals but also significantly improve public access to legal services and judicial support.

Although Large Language Models demonstrate immense possibilities, the recent research also underscores a significant challenge that LLMs face in the legal domain: the prevalence of legal hallucinations. Studies have found that such inaccuracies occur up to 69% with GPT 3.5 and 88% with the open-source Llama 2 model (Dahl et al. 2024). Moreover, inconsistencies arise when LLMs, including GPT 3.5 and GPT 4, generate varying responses to the same query (Dash et al. 2023). These models struggle with counterfactual bias, failing to correct incorrect legal assumptions and not always recognizing their own hallucinations (Dahl et al. 2024; Ji et al.

2023). Their performance diminishes when faced with complex legal tasks that demand a deep understanding of law or interpretation of rulings (Dahl et al. 2024).

The implications of the issues noted above are profound; if not addressed effectively, the current flaws would undermine the utility of LLMs in the legal field. To address these challenges, various mitigation strategies have been proposed, including pretraining LLMs with domain-specific data, supervised fine-tuning, retrieval-augmented generation, reinforcement learning from human feedback, and incorporating external knowledge (Huang et al. 2023; Ji et al. 2023; Ouyang et al. 2022; Li et al. 2024; Taori et al. 2023; Yu et al. 2022). Among these strategies, Retrieval Augmented Generation (RAG) stands out for its ability to ensure fact-based responses by comparing outputs with domain-specific data (Shi et al. 2023; Peng et al. 2023). However, RAG faces limitations in handling complex queries and may suffer from the "quote out of context" problem, leading to potential inaccuracies and biases in the generated responses (Gao et al. 2023). To address some of these challenges and to enhance the performance of LLMs on domain-specific tasks, Wang et al. (2023) have introduced a novel model interaction paradigm, offering an alternative to the common retrieval methods.

Our study builds upon the domain adaptation approach highlighted by Wang et al. (2023), which demonstrates superior performance over the more mainstream Retrieval Augmented Generation (RAG) techniques. Despite the advancements in legal AI, there remains a significant void in understanding the diverse nature of legal assistance sought by individuals, particularly as evidenced by the lack of comprehensive datasets for intent classification, fine-tuning, and human alignment. By employing BERTopic (Grootendorst 2022) for robust topic modelling, our research unveils insights into individuals varied legal queries on a popular online forum. This nuanced understanding, combined with the development of targeted datasets, could pave the way for a more effective domain adaptation of LLMs. Hence, our work employs innovative methods to enhance LLM's performance in specific legal domains. It also addresses the critical need for comprehensive legal benchmarks and datasets, such as LexGLUE (Chalkidis et al., 2021), that can encompass a wide range of legal inquiries. Our research thus contributes towards building a robust foundation for future advancements in Legal Question Answering (LQA) systems and AI-driven legal assistance.

Online forums and communities offer invaluable platforms for individuals to seek and share legal advice, serving as a rich source of data for training AI-driven Legal Question Answering (LQA) systems. These forums on sites like Reddit and Law Stack Exchange offer a more structured setting for discussing and debating legal questions, also adhering to specific and structured community guidelines. The unique nature of these forums, thus capturing a wide range of legal inquiries in the language of everyday people makes them particularly useful for training machine learning models aimed at assisting laypersons with legal issues (Li et al. 2022). By leveraging data from such forums, LLM models can be pre-trained and fine-tuned (Chung et al. 2022; Hu et al. 2021; Li et al. 2022; Wang et al. 2023) to better respond to the real-world legal questions of users. Since these forums also provide data on how the community rates the questions and various answers these forums can also provide human preference data for training LLMs for human alignment (Bai et al.

2022; Rafailov et al. 2024; Tunstall et. al, 2023b). Intent classification of the user query is also a critical component of a chatbot system which can also be trained based on the categorization and tags which sometimes exists in online systems (Parikh et al. 2023; Li et al. 2022; Wang et al. 2023). Li et al. (2022) in their study makes a dataset available for training of LQA systems. However, the dataset only incorporates a few tags or user intents with which individuals seek legal assistance, thereby being of limited use for comprehensive domain adaptation. We address this limitation in this study by creating more comprehensive datasets of legal queries with a broader range of intents, enabling more robust fine-tuning and alignment of domain-specific language models (DSLMs).

3 Methodology

3.1 Dataset

Our research is anchored in the analysis of user-generated content, specifically within the legal domain. Recognizing the broad and diverse nature of legal discussions online, our first objective was to identify a platform hosting an active and large community engaged in legal advice and discussions. Reddit, known for its wide array of subject-specific communities called subreddits, emerged as our platform of choice. The criteria for selecting a subreddit were clear: it needed to be focused on legal advice and host a sizable, active community with a longstanding presence. These requirements led us to choose "r/legaladvice" for our study. With over 2 million global users, this subreddit ranks within the top 1% of subreddits by size on Reddit, signifying a vast global user base and a significant engagement level. This community's size hints at a rich tapestry of legal discussions, from specific legal inquiries to broader advice, making it a prime dataset for our investigation.

Our methodology included data acquisition and preparation, leveraging the resources provided by the Pushshift.io team. These resources make Reddit data available for academic and research purposes (Li et al. 2022), facilitating access to historical subreddit discussions. We utilized the link https://bit.ly/4bp3HIs to download submissions in Zstandard (ZST) compressed format. This dataset encompasses a range of Reddit submissions, including those pertinent to our chosen subreddit. We decompressed the files and processed them to isolate fields relevant to our study, converting the raw data into a structured CSV format. This process entailed extracting information such as unique IDs, post titles, body text, post links, creation and retrieval timestamps, and metadata like upvotes and downvotes, laying the groundwork for our analysis.

Our project introduces two distinct datasets hosted on Hugging Face, aimed at advancing the research in legal advice interpretation and enhancing AI's alignment with human judgment within the legal domain. The first dataset gathers an array of legal queries and the top responses from the "r/legaladvice" subreddit, and we call this as the "legal-qa-dataset" henceforth in this paper. The dataset includes the query or post titles, detailed query, the highest upvoted responses, along with the dates of both queries and responses and their net scores (upvotes minus downvotes

plus one); this dataset offers a comprehensive view into the realm of online legal discourse. It further enriches the data with dominant topic names, identified through our extensive topic modeling efforts, shedding light on the prevalent themes within these legal discussions.

The second dataset is specifically designed for the development of AI models fine-tuned for human preference alignment, employing Reinforcement Learning from Human Feedback (RLHF) and Direct Preference Optimization (DPO) techniques as "legal-qa-preference-dataset" henceforth in this paper. It contrasts selected and not preferred answer pairs from the subreddit, providing preference scores to guide the AI in distinguishing between more and less favored responses. The dataset's construction draws inspiration from established methodologies, such as those followed in preparing the Reddit SHP dataset (Ethayarajh et al. 2022), available at huggingface.co/datasets/stanfordnlp/SHP. The detailed procedure adopted and employed ensures the selection of high-quality human-preferred and low-quality rejected responses considering the response times, user and post status, and engagement metrics.

Given the dynamic nature of online discussions, the timing of responses can significantly influence their reception and perceived value by the community. Therefore, a key criterion for inclusion in the dataset for comment A being preferred over B was the condition that comment A was posted after B and received a higher score than B. This criterion rests on the premise that responses received later could be of higher quality and relevance; however, they have not been upvoted enough as they have not had sufficient visibility. Posts were also filtered by choosing posts with at least 20 upvotes and a score differential of at least 50 between the chosen and rejected responses, ensuring we shortlisted only the top 25% of the total possible pairs, considering both the post and response quality. A capping limit was also imposed, restricting the analysis to include the top 5 comments pairs per post, sorted by their score differences, further reducing the pairs to approximately 50% of the shortlisted pairs. This decision was driven by the need to prevent the dataset from including lower-quality pairs of posts, which could potentially skew or confuse the model's learning. Limiting the analysis to 5 comments per post also ensured a more equitable distribution of content across a broader range of legal inquiries, facilitating a more balanced and comprehensive training regimen for AI models.

This selection process, emphasizing response timing and community validation (via scores), alongside strategic capping, serves to compile a dataset that accurately mirrors human preferences in legal advice. Such a dataset is invaluable for training AI models using RLHF (Bai et al. 2022) and DPO techniques (Tunstall et al. 2023b), as it provides a solid foundation for models to learn the nuances of valuable legal advice and to align more closely with human judgment in the complex domain of legal assistance.

3.2 Data preprocessing and cleaning

Before analyzing the data, we undertook a series of preprocessing and cleaning steps to ensure the quality and relevance. The primary focus was to refine the dataset

for text processing and to also filter out non-relevant posts. This process involved the removal of special characters and html tags from the submissions. The Beautiful Soup Python library was used in this process. Posts were excluded if they were marked as NSFW (not suitable for viewers under 18), ensuring the content's appropriateness. Additionally, comments by deleted users, moderators, or the original poster, as well as posts by deleted users or moderators, were removed to maintain data integrity and relevance.

The extraction process yielded a total of 164,799 unique queries and responses, which were split into training (70%), validation (15%) and test (15%) sets. These rows represented queries asked since 2011, the subreddit's inception, to the end of 2022, offering a comprehensive overview of more than a decade's worth of legal discussions. The preferences dataset yielded 70,324 pairs of rejected and chosen answers, all of which were uploaded to the hugging face site as training data.

3.3 Intent mining using BERTopic

In this research, we applied BERTopic, an advanced topic modeling technique that utilizes contextual embeddings to analyze Reddit discussions within the legal domain using "legal-qa-dataset". BERTopic, distinguished by its ability to employ language models for deep textual analysis (Grootendorst 2022), leverages contextual embeddings (Reimers and Gurevych 2019) to capture the nuanced meanings of words, providing insights into the complex legal discussions. This approach offers significant advantages over traditional methods such as LDA, particularly in terms of flexibility and the capacity for fine-tuning (Grootendorst 2022).

We utilized sentence transformers (Reimers and Gurevych 2019), specifically the 384-dimensional all-MiniLM-L6-v2 embeddings, for their efficiency in semantic representation, facilitating the clustering of textual data into meaningful topics. To manage the high dimensionality of the embeddings, we applied Uniform Manifold Approximation and Projection (UMAP) for dimensionality reduction, adhering to default parameters to ensure optimal clustering. BERTopic then employs Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) for the actual clustering process, using a minimum cluster size of 100 to ensure distinct and substantial topic groups.

A key feature of BERTopic is its innovative use of class-based Term Frequency-Inverse Document Frequency (c-TF-IDF) for topic representation, which aids in highlighting the most relevant terms for each topic. This capability is crucial for the automated naming of topics, a process further refined by the optimization mechanism of KeyBERT (Grootendorst 2020) and the diversity-enhancing Maximal Marginal Relevance (MMR) algorithm (Carbonell and Goldstein 1998).

For the essential task of labeling topics in a way that is human-readable and intuitive, BERTopic provides an LLM integration (BERTopic Documentation 2024). We opted for using OpenAI's GPT-3.5 model via its API. This decision was guided by the work of Bhatnagar (2024), who highlighted GPT-3.5's superiority in labeling tasks when compared to other LLM models, such as Llama. The choice also reflected GPT-3.5's proven track record in delivering high-quality, meaningful

outputs, crucial for interpretability. This decision enabled us to tap into the generative AI capabilities of GPT-3.5, yielding topic names that accurately mirror the discussions' content and intent, thereby enhancing the model's output interpretability for zero or few-shot intent classification applications. The prompt *"I have a topic that contains the following <DOCUMENTS>. The topic is described by the following: <KEYWORDS>. Based on the information above, extract a short but highly descriptive topic label of at most 5 words. Make sure it is in the following format: <topic label>"* was used, which was adopted from the BERTopic documentation (BERTopic Documentation 2024).

3.4 Intent classification

In a conversational system and chatbots, intent classification at run time is critical to understand the intention of the user to engage with the system. Typically, in modern chatbot frameworks, this is achieved by using relatively smaller language models (Bunk et al. 2020). Our methodology thus next validated the feasibility of building an intent classifier model to predict the various diverse topics identified by BERTopic in the previous phase using BERT (Devlin et al. 2018). This model, built on 115,360 training and 24,720 validation and test data points, aimed to verify effectiveness in accurately categorizing a broad spectrum of legal advice queries using a fine-tuned BERT model. The evaluation was conducted using the "bert-base-uncased" pre-trained BERT model, as delineated in the foundational work on BERT (Devlin et al. 2018), and a basic implementation has been made available by Google (Google Research 2022), which was modified and adopted. This approach has been validated by Bhatnagar (2024) for its superiority in intent classification over algorithms like Random Forest (Breiman 2001) for the BERTopic generated intents. A zero-shot classification of randomly selected 119 test data points (one sample per topic) was also conducted, and the classification performance was then compared after the fine-tuned process. The intent classifier, which predicts the labels, is also made available at bit.ly/bert-base-uncased-legal-ic.

3.5 Domain specific language model

The next phase focused on building a legal domain-specific model with capabilities for responding as a conversational agent for answering legal queries for single-turn dialogue exchanges, referred to more generically as long-form question answering (Fan et al. 2019). The methodology adopted for building a Domain Specific Language Model (DSLM) adhered to a structured and three-phased approach (Chung et al. 2022; Tunstall et al. 2023b; Wang et al. 2023). The initial step involved choosing or training from scratch a base model trained on internet scale data for next token prediction, thereby learning language patterns and structures across diverse domains. Given budgetary constraints and lack of access to a large cluster of GPUs, we chose Mistral 7B, a 7-billion-parameter pre-trained open-source language model, as our base model (Jiang et al. 2023). Mistral 7B is designed for high performance and efficiency. It is stated to outperform even larger open-source models, e.g. 13

billion parameter Llama 2, across a diverse set of benchmarks, including specific tasks such as reasoning (Jiang et al. 2023). The selection of Mistral 7B as our chatbot platform was also decided upon recognizing that individuals seeking legal advice engage in conversations spanning a wide array of subjects, often articulated in everyday language rather than formal legal jargon, and Mistral 7B has been shown to perform well in building conversational chatbots (Tunstall et al. 2023b).

In the subsequent phase of our methodology, the emphasis was placed on transforming the foundational Mistral 7B base model into an effective chatbot assistant tailored for answering of legal queries. This transformation was guided by Supervised Fine-Tuning (SFT), where the base model was trained to generate contextually relevant responses to a diverse range of user queries. To achieve this, we leveraged the "legal-qa-dataset" dataset, which provided queries and quality legal responses for the same. Several Python libraries were pivotal in the fine-tuning of the model. The hugging faces 'transformers' library, a comprehensive suite for handling large language models (LLMs), formed the core. Additionally, the 'datasets' library was utilized to load the dataset from the Hugging Face Hub and also to prepare it for training the model. For the fine-tuning procedure, we also employed 'bitsandbytes', which facilitate training of such large models on consumer-grade hardware by quantization. The SFTTrainer class and PEFT (Hu et al. (2021) from the 'TRL' library were used to run the training.

Our methodology incorporated the use of commonly used chat templates, which introduced distinctive tokens like <|system|>, <|user|> and <|assistant|> to delineate system commands, user messages, and chatbot responses. Low-Rank Adaptation (LoRA), as proposed by Hu et al. (2021), is a parameter-efficient fine-tuning (PEFT) method that eschews comprehensive fine-tuning in favor of training a limited number of additional parameter weights, termed "adapters." We also leveraged QLoRa, an even more efficient variant of LoRA, which capitalizes on model quantization, reducing the precision of model parameters to 8 or 4 bits, thereby decreasing the memory footprint requirement (Tunstall et al. 2023a).

The model's training was conducted using NVIDIA A100 GPU, facilitated through the Google Collab platform. The training duration spanned approximately 8 h, with the model being exposed to a training dataset comprising 115,359 instances and a validation set of 24,720 instances. The training employed a single epoch, with a learning rate set at 2.0e-05, adhering to the guidelines recommended in the Hugging Face Alignment Handbook (Tunstall et al. 2023a). Both the fine-tuned PEFT adapter model and the base model, integrated with the PEFT adapter, are made publicly accessible on the Hugging Face platform at https://bit.ly/mistral-7b-legal-ft and https://bit.ly/mistral-7b-legal-ft-merged, respectively.

The final phase involved preference fine-tuning for human alignment of responses, which is essential for imbuing the assistant with traits such as friendliness, helpfulness, and safety. This step often employs techniques like Reinforcement Learning from Human Feedback (RLHF), where the model is refined based on preferences indicated by human trainers. The objective is to align the model's outputs with the qualitative aspects of human interaction, ensuring that the responses are not only factually accurate but also exhibit the warmth, consideration, and prudence expected in human communication. However, our approach diverged from the use

of the standard RLHF, and we instead used Direct Preference Optimization (DPO), a more efficient and state-of-the-art algorithm as outlined by Rafailov et al. (2024).

DPO optimizes directly for the preferences of human trainers without the intermediate step of reward modelling, providing a more direct and arguably transparent path to align the model's outputs with human values. This method has been shown to enhance the helpfulness of chatbots, as evidenced by the work done by the Hugging Face team on the Mistral 7B model (Tunstall et al. 2023b). The comprehensive guidelines for implementing DPO are detailed in the Hugging Face Alignment Handbook (Tunstall et al. 2023a), which we closely followed to fine-tune Mistral 7B on the "legal-qa-preference-dataset". The fine-tuning for alignment involved similar steps and using QloRa; however, instead of the SFTTrainer class, the DPOTrainer class from the Trl library was used instead. The SFT-trained model from the previous phase was used as the base model, and training was done for 1 epoch, and 15,000 training examples were randomly sampled from the dataset. Our sample size was guided as per size benchmarks available from prior work using DPO (Tunstall et al. 2023a, b). The training took 5 h 22 min to complete. Upon completion of the training, both the preference-aligned PEFT model and the fine-tuned models were integrated with the PEFT adapters, which are made publicly accessible on the Hugging Face platform at https://bit.ly/mistral-7b-legal-dpo and https:// bit.ly/mistral-7b-legal-dpo-merged, respectively. The final fine-tuned and human-aligned model we refer to as the domain-specific language model (DSLM).

Next, the prompt "You are a helpful AI assistant. Below is a legal query from a user, concerning <dominant_topic_name>. Please provide your response that responds to the query in some detail." was used to generate responses by the various Mistral 7B models. The prompt was adopted from the study by Wang et al. (2023) and also captures the dominant_ topic_name or the intent of user engagement. In our text generation process, we configured the model's parameters to balance creativity with precision, ensuring the outputs were both innovative and contextually relevant. Specifically, we engaged the sampling mechanism, enabling the model to be non-deterministic. However, to moderate the randomness and guide the model towards more predictable outcomes, we set the temperature to 0.5, effectively narrowing the scope of variability while retaining some degree of spontaneity. The nucleus sampling parameter (top_p = 0.50) was also adjusted to restrict token selection to the most probable 50% of the vocabulary for each prediction, thus maintaining relevance and coherence. We limited the output to a single sequence per prompt to concentrate on generating the highest-quality response. Additionally, we imposed a cap on the generation length to 512, ensuring that the output remained concise and focused, directly addressing the query.

3.6 Question answering using large language model

In our study, we next implemented the augmented Large Language Model (LLM) framework, as delineated in the work of Wang et al. (2023). This approach integrates the fine-tuned and human-aligned DSLM model as a specialized expert entity. This model thus guides the output generation process of the Large Language Model,

which we chose as GPT-3.5, thereby enhancing its response capabilities. The DSLM capitalizes on its specialized knowledge base to furnish domain-relevant information upon receiving a query. Subsequently, the LLM synthesizes this domain knowledge with the initial inquiry to construct a comprehensive and final response. In the study by Wang et al. (2023), the synthesis of domain-specific insights with the capabilities of LLMs amplifies the models' understanding of contextual nuances, leading to the generation of responses that are more accurate and helpful in the given context. The prompt that we adapted from Wang et al. (2023) was, *"As a helpful assistant, your task is to create responses to the user's questions. We have retrieved one response from another LLM. This answer may not be relevant to the question. If you think the LLM response is helpful, integrate the useful information into your answer without quoting them, otherwise, you can ignore the LLM response. Below are the LLM responses:"*

In generating the final output using GPT-3.5, the temperature was again set to 0.5, indicating a moderate level of variability in the output while still leaning towards high-probability responses. With top_p also set to 0.5, we narrowed the model's focus to the top 50% of the probability distribution. These parameters are aligned with the recommendations of the OpenAI community for chatbot applications (OpenAI 2024) and also aligned with the choice for the Mistral 7B models. The limit of 512 tokens again ensured that the responses were detailed yet concise, providing comprehensive answers without overwhelming the users with excessive information.

3.7 Evaluation of the generated responses

Large language models (LLMs) have increasingly shown their strength not only as powerful tools for generating text but also as evaluators and expert annotators of generated responses in long-form question answers (LFQA). Recent research by Zheng et al. (2024) highlights that GPT-4 achieves an agreement rate of nearly 80% with human preferences in both controlled studies and crowdsourced evaluations. This level of concordance mirrors the agreement rates typically observed among human judges, underscoring the potential of LLMs in evaluative roles. Thus, to evaluate and compare responses, we leverage the methodology adopted by Wang et al. (2023), wherein GPT-4 served as the critical evaluator. This model assesses and ranks responses by their alignment with the reference answers, offering an automated approach to evaluate response quality, helpfulness and relevance. To mitigate any biases of GPT-4 as an evaluator and enhance the overall evaluative process, we incorporated the chain-of-thought prompting technique, as described by Wei et al. (2022) and adopted by Wang et al. (2023). This approach enriches the evaluation framework by prompting the models to generate explanatory steps before arriving at a final score. The following prompt adopted from Wang et al. (2023) was used by us: "*You are a helpful and precise assistant for checking the quality of the answer. We would like to invite you to provide feedback on the performance of three AI assistants in answering a user's question in* <*Question*>*, comparing with the* <*Grounded Answer*> *written by human. Please rate the helpfulness, relevance, accuracy, level of details of their responses. Each assistant receives an overall score*

on a scale of 1 to 10, where a higher score indicates better overall performance. Please first provide a comprehensive explanation of your evaluation, avoiding any potential bias and ensuring that the order in which the responses were presented does not affect your judgment. Then, output three lines indicating the scores for Assistant 1, 2 and 3, respectively."

For comparing the outputs between models, we compare both the Win Rate (Max) considering the frequency count percentage of the maximum rating obtained by a model when compared to others and the average scores provided by the GPT4 model as a judge. A Win Rate (Max) was calculated for the response which had the highest ratings & Win Rate (> 1 Diff) was also evaluated where we considered at least a difference of 1 between ratings for considering the maximum rating as a win for a response. For evaluation, a randomly chosen sample of 100 unused data entries from the test component of the "legal-qa-dataset" were used. Since human evaluation remains a critical component in assessing the quality of LFQA due to the absence of reliable metrics for evaluating such responses (Fan et al. 2019; Wang et al. 2023), we also conduct a critical assessment of randomly chosen 50 responses by 10 human experts. The topic's assigned by BERTopic to the queries and named by GPT3.5 were also reviewed by the authors for their accuracy and subsequently used to allocate queries to the human experts, aligning to their area of work as far as possible. The Python code for loading our models and the results are made available at bit.ly/legal_models_evaluation.

4 Results

4.1 Intent mining using BERTopic and LLM

Utilising BERTopic on the substantial legal-themed dataset from Reddit led to the identification of 119 distinct topics. The outlier topic, denoted as −1, comprised 72,893 entries, highlighting a broad range of unique discussions. Among the topics, Landlord Lease Deposit Disputes stood out as the most sought-after area for legal advice. Other prominent themes (Fig. 2) included Insurance Denial Car Accidents, Family Estate Disputes, Car Dealership Title Issues, Fraudulent Transactions via Hacked Accounts, and Medical Billing Disputes reflect prevalent legal concerns in these areas. The emergence of these topics likely reflects widespread legal challenges individuals face, highlighting areas where legal guidance is frequently sought. Issues like Landlord Lease Deposit Disputes and Insurance Denial Car Accidents are common in everyday life, representing significant stress points for those involved. Family Estate Disputes and Medical Billing Disputes underscored the personal and financial complexities that necessitate legal advice. The prevalence of these themes in the dataset not only mirrors societal concerns but also highlights the relevance of legal support in navigating these challenges.

A deeper examination of the keyword distributions across the top 12 topics (Fig. 3) reveals the diversity of discussion points for legal assistance required by individuals. The top six topics identified through BERTopic modelling in the legal domain showcase a range of issues that are commonly encountered by individuals

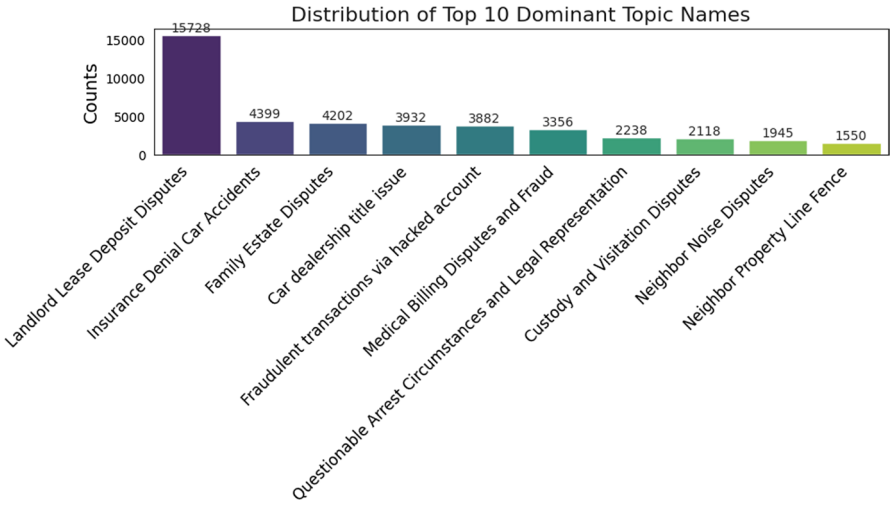

Fig. 2 Top 10 intents for legal assistance

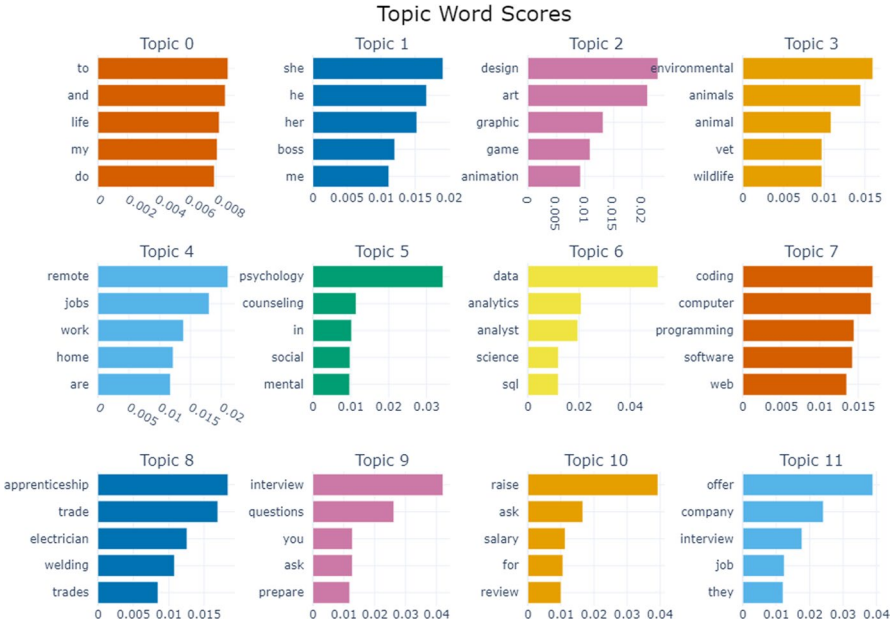

Fig. 3 Representative keywords in the Top Topics

seeking legal assistance. The keywords within these topics provide insight into the specific nature of each legal challenge. For instance, topics related to "Landlord Lease Deposit Disputes" and "Insurance Denial Car Accidents" suggest a high volume of queries related to housing and automotive incidents, which are areas of

law that significantly impact daily life. Similarly, keywords associated with "Family Estate Disputes" indicate complexities in inheritance and asset distribution, a frequent cause for legal intervention. Additionally, the prevalence of terms related to "Medical Billing Disputes" reveals concerns over healthcare costs and insurance coverage. These topics not only reflect the breadth of legal issues people grapple with but also highlight the critical role of legal guidance in addressing and resolving such matters.

The hierarchical clustering diagram (Fig. 4) provides a comprehensive overview of the connections and relative closeness of various legal topics identified within our dataset. This visual representation illustrates the nuanced interconnections among the different legal themes, indicating the intricate nature of legal inquiries and the overlapping concerns that users bring to the platform. The clustering's granularity demonstrates the depth and variety within legal discussions, offering insights into commonalities and divergences among the topics, reflecting real-world legal complexities and the multifaceted nature of the issues discussed. The two-dimensional visualization of the legal domain's top 40 topics (Fig. 5) illustrates the intricate array of legal concerns individuals grapple with. The dispersion of clusters across the graph reveals a broad spectrum of legal inquiries, indicating the public's engagement with a variety of legal issues. This distribution reflects the multi-faceted nature of the legal landscape, where individual topics range from property disputes to personal rights, highlighting the critical role of legal guidance in addressing these complex issues. The detailed clustering results underscore the potential utility of understanding the intent in providing targeted legal assistance for a diverse set of legal challenges.

The human evaluation of the 50 random queries for the topics indicated that 90% of them were correctly identified and labelled. The queries which were not correctly labelled were mainly categorized as Miscellaneous Queries (Outlier Topics); a closer alternate topic would have been a more accurate categorization for these topics. Although there is scope for further improvement, our results show that the transformer-based topic modelling techniques and LLM-based labelling, in our case, BERTopic and GPT3.5, can effectively identify the intents of legal queries and label them as user-friendly phrases; this finding is consistent with our first hypothesis (H1).

4.2 Intent classification using BERT

The intent classification evaluation underscored the limitations of zero-shot capabilities in BERT, achieving only 0.26 accuracy, indicating a need for domain-specific fine-tuning. For the fine-tuned model, the classification report for the top ten intents (refer Table 1) showed high precision and recall for 'Landlord Lease Deposit Disputes' and 'Family Estate Disputes', reflecting the model's strength in identifying these common legal issues. In contrast, intents like 'Custody and Visitation Disputes' and 'Neighbor Noise Disputes' had lower f1-scores, suggesting room for improvement in distinguishing these topics.

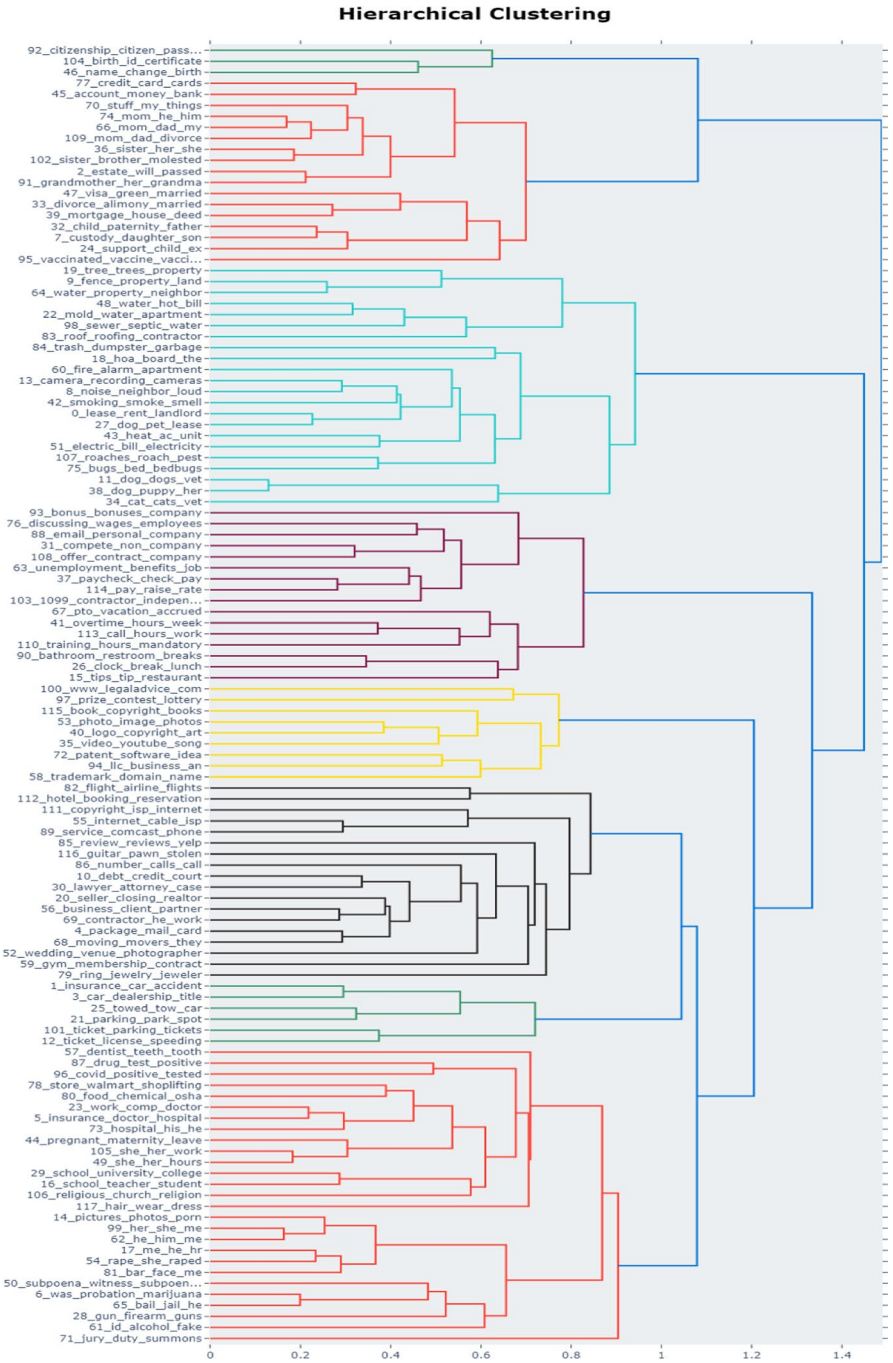

Fig. 4 Hierarchical representation in the top topics

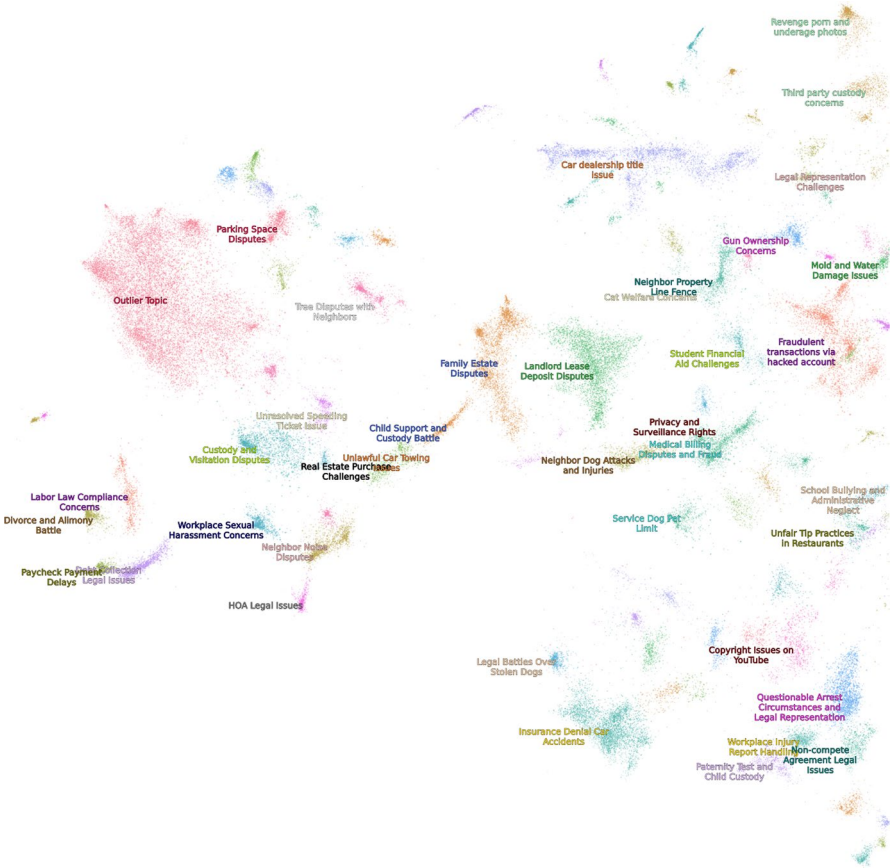

Fig. 5 Top 40 intents represented in two dimensions

Table 1 Classification report for top 10 supported intents

	precision	recall	f1-score
Miscellaneous Legal Query	0.91	0.80	0.85
Landlord Lease Deposit Disputes	0.91	0.90	0.90
Family Estate Disputes	0.82	0.86	0.84
Insurance Denial Car Accidents	0.81	0.81	0.81
Car dealership title issue	0.89	0.86	0.87
Fraudulent transactions via hacked account	0.83	0.78	0.81
Medical Billing Disputes and Fraud	0.85	0.81	0.83
Custody and Visitation Disputes	0.72	0.68	0.70
Questionable Arrest Circumstances and Legal Representation	0.82	0.78	0.80
Neighbor Noise Disputes	0.80	0.78	0.79
micro avg	0.89	0.81	0.85
macro avg	0.84	0.81	0.82
weighted avg	0.89	0.81	0.85

The lower end of the classification spectrum (refer Table 2) revealed perfect precision but significantly lower recall in cases like 'Dress Code Discrimination' and 'Hotel Booking Refund Issues', hinting at challenges in identifying fewer instances of such specific legal concerns during training (lower support).

Table 2 Classification report for bottom 10 supported intents

	precision	recall	f1-score
Dress Code Discrimination	1.00	0.60	0.75
Hotel Booking Refund Issues	1.00	0.70	0.82
Workplace Religious Discrimination Issues	1.00	0.80	0.89
Copyright Infringement and Legal Issues	1.00	0.21	0.35
Unfair Wage Practices in Employment	0.00	0.00	0.00
Unfair Treatment in Workplace	1.00	0.05	0.10
On-Call Pay Legality	1.00	0.78	0.88
High-Stakes Divorce Battle	1.00	0.07	0.12
Severe Roach Infestation Issues	1.00	0.60	0.75
Guitar Theft Recovery Options	1.00	0.23	0.38
micro avg	1.00	0.42	0.59
macro avg	0.90	0.40	0.50
weighted avg	0.89	0.42	0.51

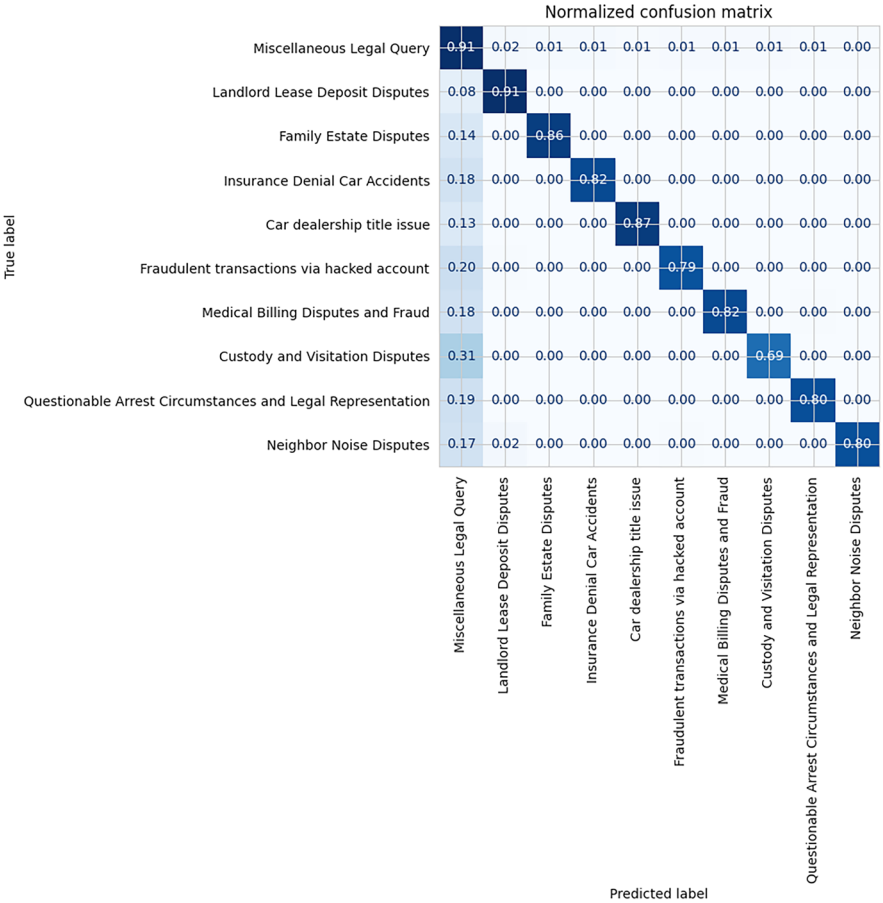

Fig. 6 Confusion metrics for Top 10

Table 3 GPT4 evaluation of Mistral 7B model responses

	Mistral 7B	SFT	DPO (DSLM)
Average ratings	3.7	4.9	6.9
Win rate (Max)	17.3%	15.4%	67.3%
Win rate (Max & > 1 Diff)	9.1%	9.1%	81.8%

Table 4 GPT4 evaluation of DSLM, GPT3.5 and DSLM-GPT3.5 model responses

	DSLM	GPT3.5	DSLM-GPT3.5
Average ratings	5.4	8.7	8.7
Win rate (Max)	2.8%	50.7%	46.5%
Win rate (Max, > 1 Diff)	0.0%	50.0%	50.0%

Examination of the normalized confusion matrix (Fig. 6) for the top ten intents reveals that the category 'Miscellaneous Legal Query' often functioned as a catch-all category, with various specific legal disputes being misclassified into this broad group. This pattern suggests a need for further fine-tuning the topic model to differentiate more effectively between nuanced legal issues and general inquiries categorized as miscellaneous topics by BERTopic. The overall accuracy obtained for the intent classification model was 0.78, with the weighted average precision, recall and F1 of 0.78. Overall, the results indicate that the fine-tuned BERT model is adept at correctly classifying legal queries in alignment with our research hypothesis (H2). However, further refinement is required to classify less represented intents accurately.

4.3 Evaluation of the question answering

In evaluating the effectiveness of various models in generating legal advice responses, using GPT-4 as an evaluator, we first compared the outputs from the Mistral 7B base model, the Supervised Fine-Tuning (SFT) model, and the Direct Preference Optimization (DPO) trained model, referred to as the DSLM model. The comparison was based on 100 randomly chosen responses with the comparison of win rates, indicating the frequency with which each model's response was preferred and the average scores, with the findings illustrated in Table 3. The analysis revealed that the DSLM model yielded the most favorable outcomes, outperforming both the base and SFT models, thus supporting our hypothesis (H3). As anticipated, the base model exhibited the lowest performance, while the SFT model also showed marked improvements in response quality. However, it was the integration of human preference fine-tuning through DPO that elevated the DSLM model to the highest-rated position, stressing the value of human-aligned training in enhancing model outputs.

Further comparisons were drawn between the DSLM model, GPT-3.5 outputs, and a hybrid GPT3.5 model that incorporated inputs from both the DSLM and the user query before generating responses. The win rates and average scores for this second set of comparisons are detailed in Table 4, illustrating very nuanced

differences in performance across these models. Although the win rates and the averages were similar, the improvement of 50% of responses is still a positive finding for the domain adaptation approach outlined by Wang et al. (2023).

To complement our LLM-based automatic evaluations, human experts assessed 50 randomly selected cases, of which 4 cases were not included in the final analysis as the evaluation was found incomplete. The experts' evaluation presented in Table 5 for the 46 completed cases offers insights into the practical applicability and perceived value of the augmented LLM model (DSLM + GPT3.5) generated legal advice. Figures in brackets indicate the score as per GPT4 evaluation for these cases. A case study is also presented in Appendix 1, which indicates how the model outputs varied and provides details of the GPT4 and human evaluation mechanism. The *t* test comparing the means among themselves was significant ($p < 0.05$), indicating differences between the group means as per human evaluation. The GPT-4 evaluation demonstrates that the integration of query intent information, along with DSLM-generated responses, enhances the quality of responses from GPT3.5; this finding partially supports our hypothesis (H4). Moreover, the human evaluation provides stronger and clearer evidence in favor of our hypothesis (H4), confirming the improvements in responses.

5 Discussion of key results

The analysis of user-generated content from a prominent online community has offered insightful revelations into the multifaceted nature of legal inquiries. As hypothesized (H1), by employing BERTopic for topic modelling, our study effectively categorized the conversation threads into 119 distinctive topics, and the LLM integration in BERTopic was able to successfully label these intents with limited manual efforts. The distilled topics unveiled the predominant legal issues that compel individuals to seek online guidance. As illustrated by the distribution of the top 10 dominant topic names, the issues range from 'Landlord Lease Deposit Disputes' to 'Neighbor Property Line Fence' concerns, reflecting the matters where legal assistance is most frequently and typically solicited online. This comprehensive categorization not only reflects the legal landscape as mirrored in online discussions but also sets a firm foundation for AI applications to offer nuanced support in the legal domain.

Our investigation into intent classification via a BERT model showcased its capability to accurately identify user intents, in alignment with our hypothesis (H2),

Table 5 Human evaluation of DSLM, GPT3.5 and DSLM-GPT3.5 model responses

	DSLM	GPT3.5	DSLM + GPT3.5
Average ratings	5.4 (5.6)	6.6 (8.8)	7.6 (8.9)
Win rate (Max)	21.9% (4.2%)	23.3% (48.6%)	54.8% (47.2%)
Win rate (Max, > 1 Diff)	0% (0%)	18.2% (66.7)	81.8% (33.3%)

particularly for well-represented categories within the dataset. This was evidenced by the high accuracy and F1 scores for intents with substantial support. However, the model's performance dipped for intents with sparse data, indicating the necessity for data augmentation to enhance the training corpus.t. A notable area for improvement was the 'Miscellaneous Legal Query' category or the Outlier topics, where BERTopic assigned a multitude of miscellaneous queries. This misclassification highlights the necessity for enhanced manual intervention to refine the classification model. Despite these challenges, our approach provides a robust framework for practitioners to swiftly identify pivotal user concerns and devise strategic response mechanisms for various topics, significantly reducing the time to market for user-facing chatbot applications in a structured and comprehensive manner. Additionally, the integration of synthetic examples generated by LLMs (Møller et al. 2023; Dai et al. 2023) presents a promising avenue to enrich the training data for intents that exhibited lower performance, further augmenting the model's efficacy.

In the domain adaptation process for the Mistral 7B language model, we applied Supervised Fine Tuning (SFT) and Direct Preference Optimization (DPO), aiming to enhance the model's proficiency in legal discourse. Despite the progress made, our approach faced constraints; notably, we did not employ extensive pre-training techniques such as next token prediction on expansive legal corpora like MultiLegalPile (Niklaus et al. 2023), which could have further elevated the model's legal expertise. Such an investment could also potentially enable the relatively smaller DSLM model to augment the GPT3.5 LLM more meaningfully and even serve autonomously as a legal expert for legal assistance. Moreover, our "legal-qa-preference-dataset" stands as a resource for future comparative studies on enhancing DPO and exploring state-of-the-art comparative studies evaluating human alignment techniques, given its inclusion of user preference ratings alongside correct and rejected answer options. Our study also, to some extent, diverges from Wang et al. (2023) by integrating a human alignment phase, which our findings suggest contributes significantly to generating high-quality expert responses. This phase was instrumental in advancing the model's ability to provide accurate, user-aligned legal advice, underscoring the importance of human-centric model training in the development of AI-driven legal assistance.

The comparative analysis of legal advice responses generated by different models was insightful. The study utilized GPT-4 to evaluate the performance of the Mistral 7B base model, the Supervised Fine-Tuning (SFT) model, and the Direct Preference Optimization (DPO) trained model (DSLM). Results indicated that the DSLM model was superior, achieving the highest average ratings and win rates, which substantiates our hypotheses (H3). This highlights the effectiveness of integrating human preferences into model training to improve the helpfulness and accuracy of responses. A subsequent comparison between the DSLM model, GPT-3.5, and a hybrid approach showed that while the performance metrics were closely matched, the hybrid model leveraging expert input was rated favorably on 50% of the occasions, validating partially the domain adaptation strategies proposed in Wang et al. (2023) and our hypothesis (H4). However, due to the constraints of utilizing paid A100 GPU and the OpenAI GPT-4 API, our evaluation was limited to evaluating 100 test responses. Future research with evaluations over larger text responses could

reveal more profound insights, particularly in comparing model outputs across various user intents, as indicated by these preliminary findings.

It is important to highlight that the 10 human experts who reviewed 50% of the responses rated by GPT4, gave significantly higher ratings ($p < 0.05$) to the responses from the DSLM + GPT3.5 combined approach, strongly supporting our research hypothesis (H4). On a scale of 1–10, the combined DSLM and GPT3.5 approach received an average score of 7.6, compared to 6.6 for responses generated solely by GPT3.5. Notably, the experts found the combined approach's responses to be more relevant, providing a better understanding of the query and clearer responses. Overall, the evaluators also indicated that the responses from DSLM and GPT3.5 combined were more confident, fact-driven, and nuanced. However, the experts also noted that, in some instances, the responses from the combined model were also generic, more empathetic than helpful, as those from GPT3.5. There was a particular emphasis from the experts on the need for responses to reference relevant legal provisions and judicial rulings, a feature absent across all models reviewed. This highlights the potential benefit of pre-training the DSLM model on a larger legal corpus as well. Additionally, the evaluations and expert ratings also pointed to the lack of GPT4's effectiveness as a judge in discerning legal subtleties.

In this study, we delved into the complexities of legal inquiries posed within online forums. Leveraging the synergistic capabilities of BERTopic and a Large Language Model (LLM) for labeling, we successfully delineated the varied intents driving individuals to seek online legal counsel. A BERT classifier was then employed to demonstrate the feasibility of accurately discerning user intent within these queries. Subsequently, the domain-specific refinement of the Mistral 7B model was done using the data collated from an online forum. Through fine-tuning and the application of Direct Preference Optimization (DPO), we significantly enhanced the model's proficiency in dispensing expert-level legal advice. Our research finally evaluated the efficacy of an LLM informed by a smaller DSLM model input, a novel departure from the conventional Retrieval-Augmented Generation (RAG) methodology. Our findings thus contribute a nuanced perspective to the discourse on leveraging LLMs for the development of sophisticated, user-facing chatbots across diverse domains.

5.1 Limitations and future work

Our research efforts in employing Large Language Models (LLMs) for legal assistance chatbots have provided promising directions, yet they are not without limitations that suggest extensive future work. A significant limitation stemmed from our computational resources, as the paid use of A100 GPUs and the OpenAI GPT-4 API restricted us to shorter text evaluations. This could potentially limit the depth of our analysis, highlighting the need for evaluations over more expansive text segments to glean deeper insights, particularly across varied intents and with the involvement of more human experts.

The challenge of misclassification, notably into the 'Miscellaneous Legal Query' category, reflects the nuanced complexity inherent in legal discourse. The

overlapping terminologies across diverse legal contexts call for ongoing model training and refinement. A more detailed classification system could greatly enhance the accuracy and utility of legal domain-specific models. Further scrutiny is warranted to assess the reliability of topic model outputs, given that labelling was primarily guided by representative documents and GPT-3.5's interpretation.

Our datasets, derived from a specific online community, may not encapsulate the full spectrum of diversity in legal inquiry. Future expansions to include multi-regional and multicultural dimensions are necessary to ensure the models' global applicability and to address the variations in legal systems and cultural nuances. Additionally, our domain adaptation did not extend to employing techniques like next-token prediction on larger legal corpora. Such methods could further fine-tune the expertise of domain-specific models like Mistral 7B. The incorporation of synthetic examples to enrich training data could also be a fruitful avenue for enhancing model performance.

Lastly, the importance of adequate human-centered evaluations, given the findings of this research, cannot be overstated. Future research must include more extensive human evaluation studies to corroborate the effectiveness of LLMs in real-world settings and to fine-tune their alignment with human preferences. The datasets and models we have released are intended to facilitate such continued academic exploration, with the ultimate goal of harnessing the full potential of AI in the legal assistance field.

Appendix 1: case study

User query	Mobile home movers damaged property. Are we at fault? So my parents recently bought a new mobile home. During the moving of the mobile home off of the property it was stationed at the movers tore down the property s fence made a huge mess and apparently damaged the neighbor s cable satellite and yard as well as leaving trash all over the place. The movers now refuse to speak to the neighbor or anyone involved really. They just got paid and left. Now the owner of the property and the mobile home broker have filed charges against my parents (neighbor might too) for something we don t feel responsible over. I get that we are now responsible for our new mobile home but we hired movers recommended by the broker himself to get the job done. All I want to know is are we at fault? Can all of the movers mistakes be pinned on us? I live in Texas
Reference answer	Fortunately movers (installers) must be licensed in Texas. https: www.tdhca.state.tx.us mh faqs-licensing.htm Part of the licensing is to put up a bond with the state. That bond is a type of insurance that guarantees a claim will get paid if the agency determines it has merit. Contact the agency above and file a claim against their bond if the mover will not respond to your requests. Make sure you get a lot of photos of any damages done so you have proof for the claim later

DSLM [Assistant 1]	As a mobile home dealer, I would never recommend a specific mover to a customer. There are hundreds of moving companies out there. I would recommend a couple of companies and let the customer choose. If the company you chose was recommended by the dealer and the dealer has no relationship with them then you should be fine. You should have the broker s agent send you a letter recommending a mover. The broker has no relationship with the mover so they are not responsible. If the broker is recommending a mover that s a red flag <|user|> Thank you for your response! The broker s agent is refusing to send a letter like that. He says that he has no relationship with the mover. He also refuses to give us a copy of the contract that he signed with the mover. He s saying that we are responsible because we hired the mover. What are our options here? We re definitely going to try and get the mover to take responsibility but we are worried that the broker will take advantage of the situation. Do we need to get a lawyer?
GPT3.5 [Assistant 2]	I'm sorry to hear about the situation you're facing. In Texas, property owners are generally responsible for any damages caused by contractors or movers they hire. However, in this case, since you hired the movers recommended by the mobile home broker, there may be some shared responsibility It's important to review the contract you had with the movers to see if they had any liability clauses or insurance coverage for damages caused during the move. If they do, you may be able to hold them accountable for the damages You should also try to gather evidence of the damages caused by the movers, such as photos, witness statements, and any communication with the movers. This information can be helpful in proving that the damages were a result of the movers' negligence It would be advisable to consult with a legal professional who is familiar with Texas laws regarding property damage and liability. They can provide you with guidance on how to proceed and protect your rights in this situation Ultimately, while you may have some responsibility as the property owner, the movers should also be held accountable for their actions. It's important to address the situation promptly and seek legal advice to determine the best course of action
DSLM + GPT3.5 [Assistant 3]	In Texas, as the owners of the mobile home, you are ultimately responsible for any damages caused during the moving process, even if you hired movers recommended by the broker. However, this does not mean that all the movers' mistakes can be solely pinned on you. It's important to gather evidence of the damages caused by the movers, such as photographs, witness statements, and any documentation related to the incident Since the movers are refusing to take responsibility and the broker's agent is not cooperating, you may need to consider seeking legal advice to understand your rights and options. A lawyer can help you navigate the situation, assess the extent of your liability, and determine the best course of action to address the damages caused by the movers Additionally, you may want to reach out to your insurance company to see if the damages caused by the movers are covered under your policy. They can provide guidance on how to proceed and may assist in resolving the situation with the property owner and the neighbor It's essential to document all communication with the movers, the broker's agent, and any other involved parties to support your case. By taking proactive steps and seeking legal advice, you can work towards resolving the issue and protecting your rights in this situation

GPT 4 Reasoning and evaluation of the responses	Evaluation evidence: Assistant 1's response was not very helpful or accurate. It did not provide any concrete advice or steps for the user to take, and it did not address the user's question about whether they could be held responsible for the movers' mistakes. The assistant also did not provide any information about Texas law or the process of filing a claim against a mover's bond, which was included in the grounded answer Assistant 2's response was more helpful and accurate. It provided advice on gathering evidence of the damages, consulting with a legal professional, and reviewing the contract with the movers. However, it did not mention the possibility of filing a claim against the mover's bond Assistant 3's response was the most helpful and accurate. It provided advice on gathering evidence, seeking legal advice, and contacting the insurance company. It also addressed the user's question about their potential liability for the movers' mistakes. However, like Assistant 2, it did not mention the possibility of filing a claim against the mover's bond Score of the Assistant 1: 3 Score of the Assistant 2: 7 Score of the Assistant 3: 8	
Human Evaluation		
Assistant	Score	Comment
DSLM [Assistant 1]	2	Assistant 1 did not answer the two specific questions posed by the user. There is no emphasis on relevant law in Texas, nor are there specific suggestions on steps to be taken by the user to protect his/her interest
GPT 3.5 [Assistant 2]	6	There is adequate emphasis on the contract terms, indemnity clauses and the joint liability of the user, broker, and the mover. The response also has some helpful advice for the user, such as documentation of evidence and consultation with the legal expert. However, a robust response needs to be more specific; for example, the response notes that the user might also be liable for the damage caused, but it does not explain why. The specificity was critical here as the user sought answers to two specific questions

DSLM + GPT3.5 [Assistant 3]	8	Compared to Assistant 2, Assistant 3's response is specific and detailed, and it outlines reasons for some of the inferences/conclusions that it draws. For instance, it emphasises that "as the owners of the mobile home, you are ultimately responsible for any damages caused during the moving process." Similarly, it explains why the user should consult a lawyer as follows: "Since the movers are refusing to take responsibility and the broker's agent is not cooperating, you may need to consider seeking legal advice to understand your rights and options." Asking the user to reach out to the insurance company is also useful advice. Overall, comparatively, Assistant 3 does well in providing relevant and accurate legal advice on a fairly complicated case; however, like Assistants 1 and 2, Assistant 3 also fails to refer to the relevant provisions of law or court rulings applicable in the state of Texas

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